Trajectory Optimization in Vector Flow: Solving the Swimmer's Problem with Numerical Methods

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Abstract

In this work, the optimal path-planning problem for a swimmer crossing a river in the presence of downstream flow is investigated through mathematical modeling, numerical analysis, and optimization techniques. Several physical scenarios are examined, including spatially uniform river flow with both constant and exponentially decreasing swimmer velocity, the latter being introduced to model swimmer fatigue over time. The influence of key environmental and physical parameters—such as river width, swimmer speed, downstream current intensity, and fatigue rate is systematically analyzed to evaluate their impact on the resulting trajectories and crossing times. In addition, a more advanced movement model involving non-uniform spatially distributed flow fields is developed. This case is numerically treated using the finite-difference method to discretize the governing equations and formulate a trajectory optimization problem. The optimal time-dependent sequence of swimming angles is then determined through a gradient-descent–based optimization algorithm, enabling the swimmer to adapt dynamically to spatial changes in the flow field. The project demonstrates how concepts from fluid dynamics, physics, calculus, numerical methods, and optimization can be combined to study trajectory planning in complex environments. More broadly, it highlights the applicability of computational and physics-based approaches to real-world dynamical systems and optimal-control problems.

KEYWORDS

Vector Flow, Finite Difference, Gradient Descent Optimization

1. Introduction

Consider the task of a swimmer crossing a river, with the river exhibiting arbitrary stationary flow. Given that the swimmer has control of the swimming direction, what is the optimal swimming strategy (e.g. swimming angle time-series), in order to reach a target point in the opposite bank as closely as possible? A graphical depiction is provided in Fig. 1. Therefore, given a flow field across the river banks, as well as a starting (A) and ending (B) point, the goal is to identify an optimal strategy that the swimmer can attempt in order to cross the river $A \rightarrow B$.

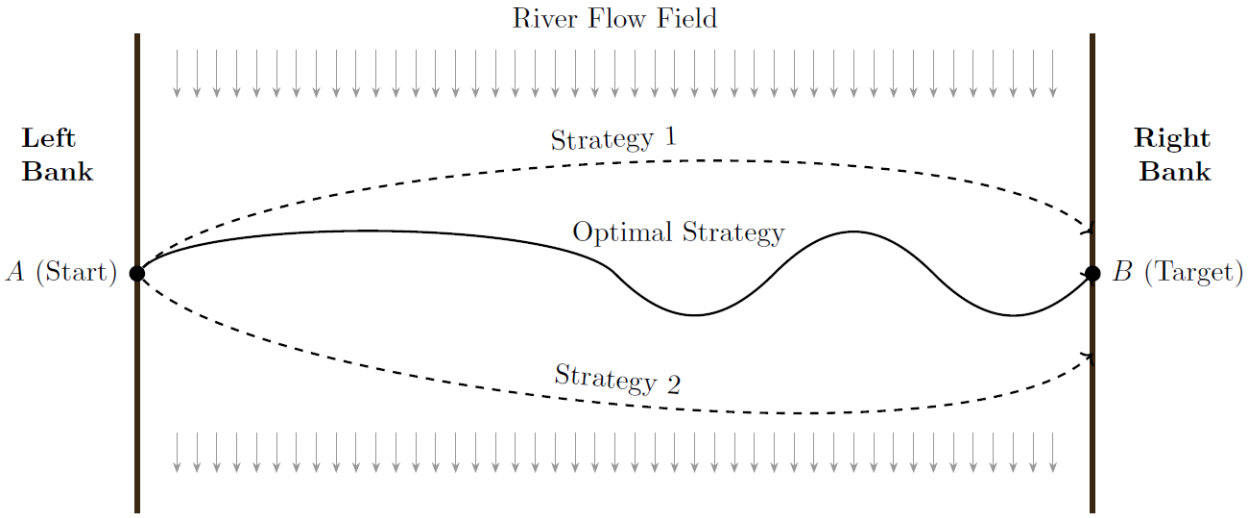


Figure 1: Graphical demonstration of the river crossing concept described.

In pertinent literature, several studies can be found that make use of various optimization approaches on the path-planning problem on liquid surfaces. In (1), an improved particle swam optimization (PSO) approach, which exploits an adaptive information dynamic adjustment strategy, was developed to enhance the optimal path identification of a water surface garbage cleaning robot. Another PSO approach was developed in (2) to optimize navigation time and battery efficiency for Unmanned Surface Vehicle (USV). In (3), a modified Genetic Algorithm that includes a stretched fitness function, an adaptive mutation and a crossover operator, was introduced to optimize the Complete Coverage Path Planning (CCPP) problem of a USV on complex and irregular boundaries. In addition, (4) describes an adversarial-reinforcement learning strategy that assists microswimmers to find optimal paths in in turbulent flows.

In this work, the case of uniform river downstream with constant and exponentially decreasing swimmer velocity as well as the case of a spatially-dependent river downstream with constant swimmer's absolute speed, are analyzed. The developed methodology relies on the simulated model's physics but exploits the numerical scheme of finite differences to solve the flow-speed differential equations and the gradient descent optimization scheme to reveal the optimal swimming path in the case of spatially unstable flows.

In fact, the case of anomalies in potential stream flows has been examined in (5) where the Rankine oval geometry is analyzed, as well as in (6), where advanced machine learning models are employed to predict them on a river. Another study is presented in (7), where data from RADARSTAT and CNNs are used to estimate the river flow. Furthermore, in (8) a finite difference method and a modified Gradient Descent Algorithm (GDA) were also studied on the fluid flow problem over a linearly stretching surface.

In this particular project, however, the scope is to present a simplified approach to facilitate a general understanding of the problem before proceeding to the research of more advanced methodologies. In this way this project aims to set a basic research framework that will act, primarily, as a tutorial. The rest of the report is organized as follows. In Section 2, the methodology of tracking the swimmer's path in the presence of a constant uniform river flow is analyzed,

assuming a) constant swimmer's speed, b) exponentially decreasing swimmer's speed and c) spatially distributed river flow. Furthermore, in Section 3 the parametrization of each case's simulations and the corresponding results are demonstrated. Finally, section 4 concludes this work.

2. Methods and Materials

In this work, three different case studies according to a number of physical conditions and assumptions are considered. In each one, the physics-mathematical model is established, while numerical techniques have been employed to maintain the corresponding optimal solution for the swimmer's trajectory to reach the target point. The cases are described as follows: a) constant and uniform river flow with constant swimmer's speed, b) constant and uniform river flow with exponentially decreasing swimmer's speed and c) non-uniform river flow with constant swimmer's speed.

2.1 Constant River flow and swimmer's speed

In the first case study, we assume a constant and uniform river current flow of downstream speed u_s (m/s), where the swimmer also has a constant speed $\boldsymbol{u_m}$ (m/s) relative to the river, starting from the point $\boldsymbol{r_A} = [x_A \quad y_A]^T$. The river downstream can be expressed in vector form as $\boldsymbol{u_s} = [0 \quad -u_s]^T$, while the relative speed of the swimmer is $u_m = [u_m \cos\theta \quad u_m \sin\theta]^T$. In this expression, θ is the angle of movement of the swimmer from the x-axis. Modeling the interaction of the swimmer's instantaneous velocity with the rivers flow as linear, the swimmer's resulting speed (relative to the ground) is:

$$\boldsymbol{u_t} = \boldsymbol{u_s} + \boldsymbol{u_m} = \begin{bmatrix} u_m \cos\theta \\ u_m \sin\theta - u_s \end{bmatrix} \quad \text{(eq1)}$$

In this case, the problem is to find a constant angle θ of the swimmer's relative speed, for which a desired (target) position $r_t = [x_A + D \quad y_t]^T$ at the opposite river bank can be reached. D denotes the river's width which is assumed constant within the swimming area. To solve this problem, the total time needed for the swimmer to reach the opposite bank can be expressed as the fraction of the x-axis distance between the banks and x-axis swimmer's speed:

$$t_d = \frac{D}{u_m \cos\theta} \qquad \text{(eq2)}$$

Therefore, using (eq1) and (eq2) and since u_t is constant, the movement equation from r_A to r_t can be written as:

$$\boldsymbol{r_t} = \boldsymbol{r_A} + \boldsymbol{u_s} t_d = \begin{bmatrix} x_A + D \\ y_A + D\tan\theta - \frac{D u_s}{u_m \cos\theta} \end{bmatrix} \text{(eq3)}$$

However, some physical restrictions should be considered to exploit the model described by (eq3):

- The swimmer has a maximum possible absolute speed of $u_{m,max}$.
- The swimming relative angle θ should obey the restriction $\theta \neq (eq2k + 1)\pi$ rad, for any $k \in Z$.

- In order for the swimmer to reach the opposite bank in a realistic finite time, their relative speed and angle should meet some criteria compared to the downstream speed. Although it has already been derived that when θ gets the values ±90°, t_d tends to infinite, it has heuristically been assumed through simulations that θ should be bounded between $\theta = \pm 75^o$.
- For the same reason, we assume that the maximum absolute swimmer's speed $u_{m,max}$ should be greater, than the absolute downstream speed u_s.

From the form of (eq3), it is observed that a closed form solution for θ cannot be derived. Therefore, an optimal θ^* that minimizes the distance between the desired (target) destination r_t and the point that the swimmer will reach, defined as r_B is searched. This optimization problem can be expressed as:

$$\theta^* = argmin\{J(\theta)\}, \text{ with } J(\theta) = |\boldsymbol{r_t} - \boldsymbol{r_B}(\theta)| \quad \text{(eq4)}$$

The problem described by (eq4) is numerically solved with line search through a vector of angles $\theta = \{-75^o, -74.99^o, \ldots, 74.99^o, 75^o\}$ and (eq3), by using the Numpy library of Python programming language. A simple demonstration of this case is shown if Fig. 2.

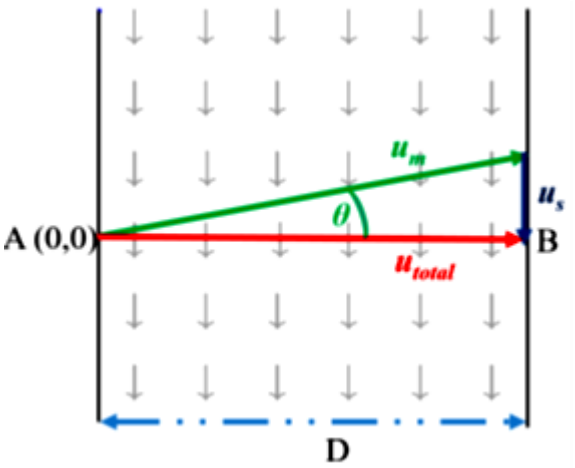


Figure 2: Constant downstream and swimmer's relative speed: Simplified sketch showing the velocities' vectors.

2.2 Constant river flow with exponentially decreasing swimmer's speed

Aiming to depict a realistic scenario, in which a swimmer would obviously get tired as time goes by, u_m is considered as an exponentially decreasing value over time. Angle θ remains as a constant that leads to a particular vertical displacement. A brief visual description of this concept that depicts the velocity vectors is shown in Fig. 3. In this case, the relative speed of the swimmer becomes:

$$\boldsymbol{u_m}(t) = u_{m0}e^{-at}[\cos\theta \quad \sin\theta]^T \quad \text{(eq5)}$$

where α>0 is the exponent decreasing factor with units of 1/sec and u_{m0} is the initial swimmer's relative speed (m/sec). Therefore, the total speed is expressed as follows:

$$\boldsymbol{u_t}(t) = [u_{m0}e^{-at}\cos\theta \quad u_{m0}e^{-at}\sin\theta - u_s]^T \qquad \text{(eq6)}$$

Since the speed is not constant, we write the general movement equation for t>0:

$$\boldsymbol{r}(t) = \boldsymbol{r_A} + \int_0^t \boldsymbol{u_t}(\tau)d\tau \qquad \text{(eq7)}$$

At $t=t_d$ the river has been crossed and the swimmer's position is:

$$\boldsymbol{r_B}(t) = \boldsymbol{r_A} + \int_0^{t_d} \boldsymbol{u_t}(\tau)d\tau \qquad \text{(eq8)}$$

The above can be written in the following vectorized form:

$$\boldsymbol{r_B}(t) = \boldsymbol{r_A} + \begin{bmatrix} \int_0^{t_d} u_{m0}e^{-a\tau}\cos\theta\, d\tau \\ \int_0^{t_d} (u_{m0}e^{-a\tau}\, sin\,\theta - u_s)d\tau \end{bmatrix} \qquad \text{(eq9)}$$

By the calculation of the x-axis element integral in (eq9) and considering that $D=x_B-x_A$ we get:

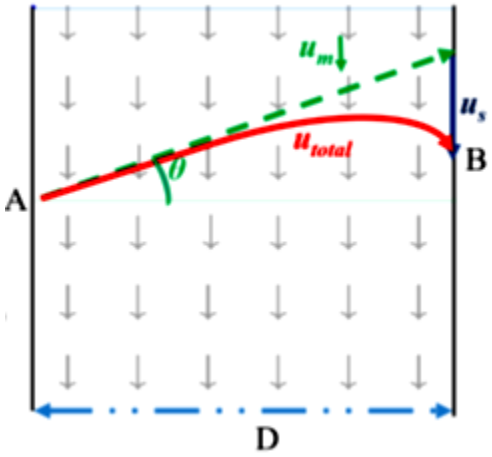


Figure 3: Constant downstream with exponentially decreasing swimmer's relative speed: Simplified sketch showing the velocities' vectors.

$$D = \frac{1}{a}u_{m0}\cos\theta\,(1 - e^{-a\tau}) \qquad \text{(eq10)}$$

From (eq10) the total crossing time t_d can be derived:

$$t_d = \frac{1}{a}\ln\left(\frac{u_{m0}\cos\theta}{u_{m0}\cos\theta - Da}\right) \qquad \text{(eq11)}$$

Furthermore, by the calculation of the y-axis element integral in (9) we get:

$$y_B = y_A + \frac{1}{a}u_{m0}\sin\theta\,(1 - e^{-a\tau}) - u_s t_d \qquad \text{(eq12)}$$

Plugging (eq11) into (eq12) the following expression for the final vertical position of the swimmer is derived:

$$y_B = y_A + 1 + \frac{u_{m0}\cos\theta}{u_{m0}\cos\theta - Da} - \frac{u_s}{a}\ln\left(\frac{u_{m0}\cos\theta}{u_{m0}\cos\theta - Da}\right) \qquad \text{(eq13)}$$

A closed-form solution for the angle θ, given a specific target for y_B cannot be derived from (eq13). Therefore, the problem is solved numerically, similarly as in the previous case, by finding a θ^* that minimizes (eq4).

2.3 Distributed river flow with constant absolute swimmer's speed

In this case, we consider a non-uniform field-dependent river current flow speed, described by a 2-dimensional spatial function $\boldsymbol{f}(x,y) = [f_x(x,y) \quad f_y(x,y)]^T$, which is time-independent. The swimmer is forced to adapt their relative speed accordingly during the river crossing time. In terms of this particular project and case, we assume that the swimmer keeps their relative absolute speed u_m constant, while adapting their swimming angle as a time function θ(t), t ∈ [0, td].

Minor disturbances in the river flow speed usually do not have a significant impact on the swimmer's orbit between the river banks. Therefore, when using a particular set of river flow parameter values, for large river widths, the disturbance can be neglected and approximate the problem solution in the same way as described in the case of uniform river flow, i.e. assuming a constant relative swimming direction θ.

However, if the river width is relatively small or the river flow disturbance is stronger. The swimmer will be forced to temporarily turn direction in order to avoid areas with non-uniform river flow (obstacles such as rocks might reserve the centers of these areas). Therefore, the relative angle of direction of the swimmer θ is not considered further as constant, but as a function of time θ(t). To simplify the approach, we assume as mentioned above that the swimmer's absolute relative speed u_m is constant. Then, the expression of the total speed of the swimmer in (eq1) is modified as follows:

$$\boldsymbol{u_t}(x,y,t) = \boldsymbol{u_m}(t) + \boldsymbol{f}(x,y) = \begin{bmatrix} u_m \cos\theta(t) + f_x(x,y) \\ u_m \sin\theta(t) + f_y(x,y) \end{bmatrix} \quad \text{(eq14)}$$

Although the previous expression and the analysis below stands for each time independent $\boldsymbol{f}(x,y)$, in terms of this work, simulations have been carried out for the case in which river flow speed vector is expressed as follows:

$$\boldsymbol{f}(x,y) = \begin{bmatrix} \frac{Q}{2\pi}\left(\frac{x-x_{m1}}{R_1^2} - \frac{x-x_{m2}}{R_2^2}\right) \\ -V + \frac{Q}{2\pi}\left(\frac{y-y_{m1}}{R_1^2} - \frac{y-y_{m2}}{R_2^2}\right) \end{bmatrix} \quad \text{(eq15)}$$

where $R_1 = \sqrt{(x-x_{m1})^2 + (y-y_{m1})^2}$ and $R_2 = \sqrt{(x-x_{m2})^2 + (y-y_{m2})^2}$.
This field actually demonstrates a source–sink pair embedded in a uniform background flow (Rankine-oval-type potential flow (9,10)). The source and sink are centered on (x_{m1}, y_{m1}) and (x_{m2}, y_{m2}) respectively, with an equal action width parameter Q, while V is the offset downstream speed. In case of Q = 0, the field is reduced to the previous cases where the stream is uniform. A visualization example of $\boldsymbol{f}$(x, y) for custom parameters' selection is demonstrated in Fig. 4, where the 2 whirlpools can be observed.

In particular, the speed vectors rotate around the points (x_{m1}, y_{m1}) and (x_{m2}, y_{m2}), changing the stream's behavior. The cause of this behavior might be the morphology of the river's bottom or a temporary disturbance of the wind direction that forms whirls. If we use a set of particular parameters for larger river widths (D), the disturbance to the river stream's uniformness will be less significant.

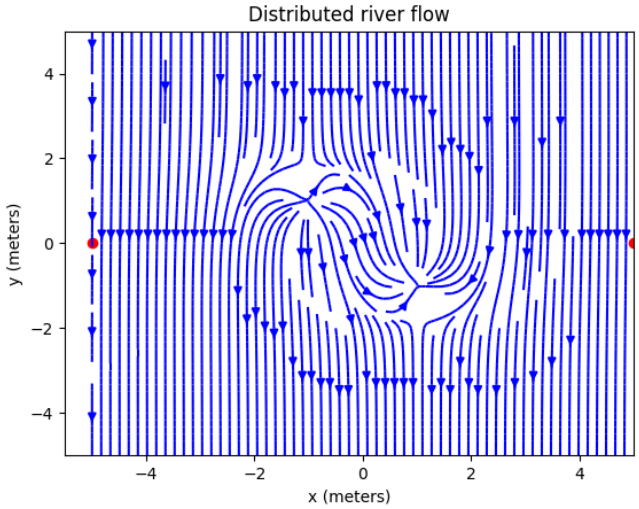


Figure 4: River flow visualization, following (eq15) for D=10m, V=1m/s, Q=1 m^2/s, y_t=5m, α=0.005, x_{m1}=-1m, y_{m1}=1m, x_{m2}=1m, y_{m2}=-1m.

For each time-independent $\boldsymbol{f}(x,y)$, it is obvious that u_t depends on the position which changes over time. Also, since the speed is not constant, the movement equation is written in the form (eq7) (modified for spatial-dependent total velocity):

$$\boldsymbol{r}(t) = \boldsymbol{r_A} + \int_0^t u_t(x(\tau), y(\tau), \tau)\, d\tau \qquad \text{(eq16)}$$

In the integral expression above, the total time needed to cross the river $t=t_d$ cannot be directly derived, as in the previous uniform downstream cases. Instead, (eq16) can be differentiated to exploit the velocity definition from physics:

$$\frac{dr(t)}{dt} = u_t(x(t), y(t), t) \qquad \text{(eq17)}$$

The total speed in (eq16) seems to be a function of both the swimmer's position and time. The swimmer's position itself changes in time, so the whole expression depends on time in a recursive way. Furthermore, as already mentioned, since the angle θ(t) in (eq14) is unknown, a clear expression of speed as a function of time cannot be analytically derived.

Taking into account the remarks above, the problem is approached numerically, using the finite difference method (FDA) (11, 12) to approximate the 1st-order derivatives in (eq16) and express the problem in a recursive form. Furthermore, an initial guess of the total time is made t_d to be properly modified until a framework for optimal θ^* can be provided. The time $t \in [0, t_d]$ should be sampled in a number of N_p samples, with a time step of $\Delta t = t_d/N_p$. The derivative expressions for x and y-axis are then written as follows:

$$x_{k+1} = x_k + u_{t,x}(x_k, y_k, \theta_k)\Delta t, \qquad k=\{0,1,\ldots,N_p\text{-}1\} \qquad \text{(eq18)}$$
$$y_{k+1} = y_k + u_{t,y}(x_k, y_k, \theta_k)\Delta t, \qquad k=\{0,1,\ldots,N_p\text{-}1\} \qquad \text{(eq19)}$$

where the following notation is used for simplicity: $x_k = x(t_k)$, $y_k = y(t_k)$, $\theta_k = \theta(t_k)$.
The following condition is also defined (initial position):

$$\boldsymbol{r_0} = (x_0, y_0) = (x_A, y_A) = \left(-\frac{D}{2}, 0\right) \quad \text{(eq20)}$$

While the target position is set at:

$$\boldsymbol{r}_t = (x_t, y_t) = \left(\frac{D}{2}, y_t\right) \qquad \text{(eq21)}$$

where y_t is the target deviation along the y-axis.

The whole problem is actually an optimization of the angle function θ(t) in such way that the swimmer approaches as close to $\mathbf{r_t}$ as possible in N_p time points (samples) with a constant defined absolute speed u_m. On each time sample t_k, the relative angle of the swimming direction will get a particular value θ_k. Therefore, an optimal sequence for the angle vector $\boldsymbol{\theta} = [\theta_0, \theta_1, ..., \theta_{Np-1}]$ is searched. This sequence corresponds to the discretized form of θ(t). This non-linear optimization problem can be solved by using the Gradient Descent Algorithm (GDA), which is set to minimize the following cost function:

$$\boldsymbol{\theta}^* = argmin\{J(\boldsymbol{\theta})\}, \text{ with } J(\boldsymbol{\theta}) = \left\| \boldsymbol{r_t} - \boldsymbol{r_{N_p}}(\boldsymbol{\theta}) \right\|^2 \quad \text{(eqx 22)}$$

Approaching a close to zero result for J($\boldsymbol{\theta}$) means that for the optimal sequence $\boldsymbol{\theta}^*$ the target $\mathbf{r_t}$ is almost accomplished. Instead of absolute values, the square form is used (least-squares approach) in (eq21) to maintain proper differentiation during the GDA steps.

2.4 Gradient Descent Optimization

The GDA is an optimization scheme that tends to optimize a set of parameters (in our case the $\boldsymbol{\theta} \in R^{Np}$ sequence) by iteratively minimizing a cost function (in our case J(θ)) (13-15). This is performed by guiding the model "downhill' to find the lowest point, moving by an opposite direction to the gradient by a learning step $\alpha > 0$. Within each step, the cost function is recalculated. If it exceeds a predefined maximum allowed error, the gradient $\nabla J_\theta \in R^{Np}$ is calculated and the parameter θ update is performed. The parameters θ are initialized to zero or a sequence of specific values at the beginning ($\boldsymbol{\theta}^{(0)} = [\theta_0^{(0)}, \theta_1^{(0)}, ..., \theta_{Np-1}^{(0)}]$).

Within this case, the cost function J($\boldsymbol{\theta}$) calculation is recursively performed by computing the next position for each time sample k. In particular, the starting point $\mathbf{r_A}$ (eq19) and the current $\boldsymbol{\theta}$ vector are replaced in (eq18) and (eq19) for k = 0. Then, through the recursions (N_p times) the final expected position $\mathbf{r_{Np}}$ is obtained and the cost is computed using (eq22). If the cost is greater than expected, its gradient is numerically estimated using the FDM (see also steps 6 and 7 in Algorithm 1 Fig. 5). The whole process is accurately described in Algorithm 1 (Fig. 5), while a brief flow diagram is demonstrated in Fig. 6. GDA is a scheme characterized by simplicity and efficiency,

scalability, and flexibility, advantages that have led to its adaptation in many machine learning applications. However, it is sensitive to the learning rate selection (α), an issue that may lead to local minima points instead of global ones at non-convex optimization problems.

Algorithm 1: Gradient Descent Process for path optimization

Data: N_p, y_t, $k = 0$, D, $\mathbf{r_o} = (x_o, y_o) = (-D/2, 0), y_t, \epsilon_{stop}, \epsilon_\theta$
Result: θ^*
1. Solve the problem for uniform river speed $\mathbf{u_s}$ and the desired y_t.
2. Acquire a constant θ_0 and estimate the time to cross the river t_{d0}.
3. Initialize total time as $t_d = 1.2t_{d0}$. Discretize t_d to N_p samples.
4. Define the initial $\theta^{(0)} = [\theta_0^{(0)}, \theta_1^{(0)}, \ldots \theta_{N_p-1}^{(0)}] \in \mathbb{R}^{N_p \times 1}$ with the values θ_0 acquired in step 1.
5. Compute the cost $J(\theta^{(0)})$, recursively by using by using (18), (19) and (22).

while $J(\theta^{(k)}) \geq \epsilon_{stop}$ **do**

6. Estimate the following costs, corresponding to each dimension of $\theta^{(k)} \in \mathbb{R}^{N_p \times 1}$:

$$J(\theta_0^{(k)} \pm \epsilon_\theta, \theta_1^{(k)}, \ldots, \theta_{N_p-1}^{(k)}),$$
$$J(\theta_0^{(k)}, \theta_1^{(k)} \pm \epsilon_\theta, \ldots, \theta_{N_p-1}^{(k)}),$$
$$\ldots$$
$$J(\theta_0^{(k)}, \theta_1^{(k)}, \ldots, \theta_{N_p-1}^{(k)} \pm \epsilon_\theta),$$

recursively by using(18), (19) and (22) (complexity $\mathcal{O}(N_p^2)$).

7. Estimate the partial derivative of each dimension of the vector $\theta^{(k)}$ using the central-scheme finite difference method:

$$\frac{\partial J}{\partial \theta_i} = \frac{J(\theta_0^{(k)}, \ldots, \theta_i^{(k)} + \epsilon_\theta, \ldots, \theta_{N_p-1}^{(k)}) - J(\theta_0^{(k)}, \ldots, \theta_i^{(k)} - \epsilon_\theta, \ldots, \theta_{N_p-1}^{(k)})}{2\epsilon_\theta}$$

8. Compute the gradient: $\nabla J_{\theta^{(k)}} = \left[\frac{\partial J}{\partial \theta_i}\right]_{i=0}^{N_p-1} \in \mathbb{R}^{N_p \times 1}$.
9. Apply the learning rule: $\theta^{(k+1)} = \theta^{(k)} - \alpha \nabla J_{\theta^{(k)}}$
10. Update $k \to k + 1$;

end

Figure 5: Description of the GDA scheme applied for the $\boldsymbol{\theta}$ sequence optimization.

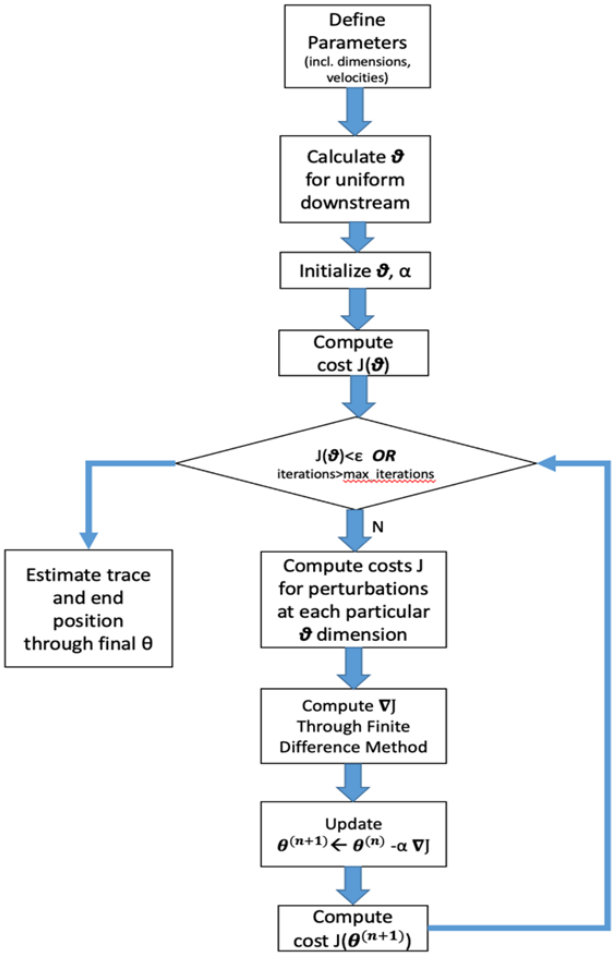


Figure 6: Simplified process flow of the GDA scheme applied.

3. Results and Discussion

In this section, a number of simulations for the testcases were carried out, considering uniform and non-uniform river flow and tuning critical parameters, such as the river's width, the swimmer's relative speed and the downstream speed to observe and validate their impact to the swimmer's actual trajectory and final point. For all the simulations, the Python programming language was used, employing the Numpy and Matplotlib libraries.

3.1 Case constant river flow and swimmer's speed

For the constant river flow case and swimmer's speed case, the final target y_t, the swimmer's absolute relative speed u_m and the river's width D were parametrized through 3 corresponding examples, while the final point vertical position y_B was extracted as a function of a constant swimming angle θ, using (3). The parameters' values for each particular example are demonstrated in Table 1.

In the first simulation example, the optimal swimming angles θ for a list of multiple y_t target positions (−200, −150, −100, −50, 0, 50, 100) m are shown in Fig. 7. In addition, the dependence between θ and the final vertical position y_B under various swimmer speeds um is demonstrated in

Fig. 8 (second simulation example), where it is observed that for larger u_m, smaller swimming angle θ is needed to eliminate vertical displacement y_B. Finally, the corresponding graph for the width of the river D as a parameter (third simulation example), is shown in Fig. 9. It is observed that the swimmer deviates from their final position target more intensively, for larger river width values D and for the same swimming angle θ. Therefore, higher values of θ are needed to decrease y_B. We can also observe that for a constant swimming angle of $\theta = 20^o$, the vertical displacement is $y_B = 0$m, independently of the width of the river D.

Table 1: Parameters settled for uniform downstream and constant swimmer's speed case simulation examples.

Parameter	**Example 1**	**Example 2**	**Example 3**
River Width D in m	100	100	{10, 20, ..., 100}
River downstream u_s (m/s)	1	1	1
Swimmer's relative speed u_m (m/s)	1.5	{1.2, 1.4, ..., 3}	1.7
Final Target Position y_t (m)	{−200, −150, ..., 100}	0	0

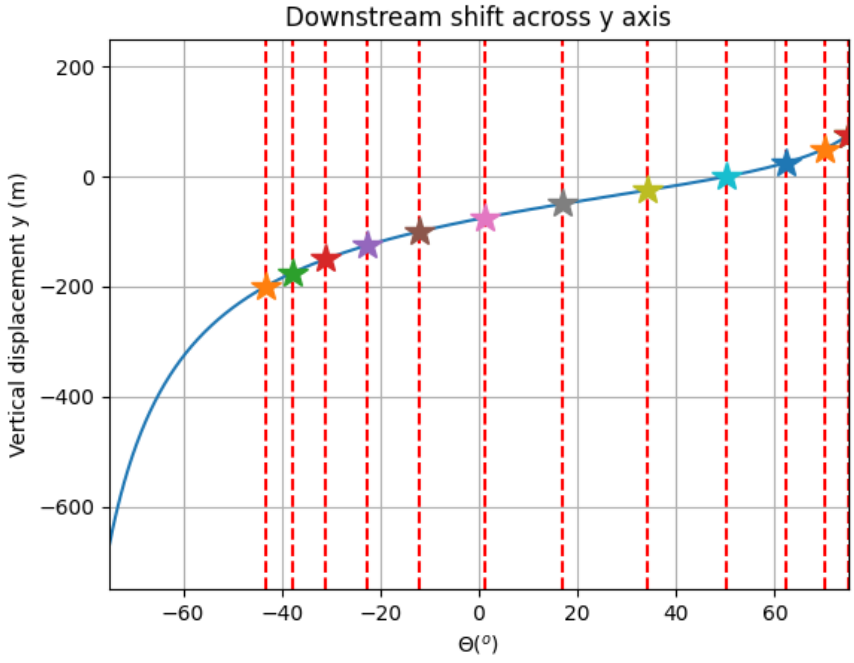


Figure 7: Dependence of the vertical displacement y_B from the swimmer's constant movement angle θ, with optimal solution for the parameter values mentioned in Table 1, example 1 (u_s = 1m/s, u_m = 1.5m/s, D = 100m, desired y_t = {−200, −175, −150, −125, −100, −75, −50, −25, 0, 25, 50, 100}m).

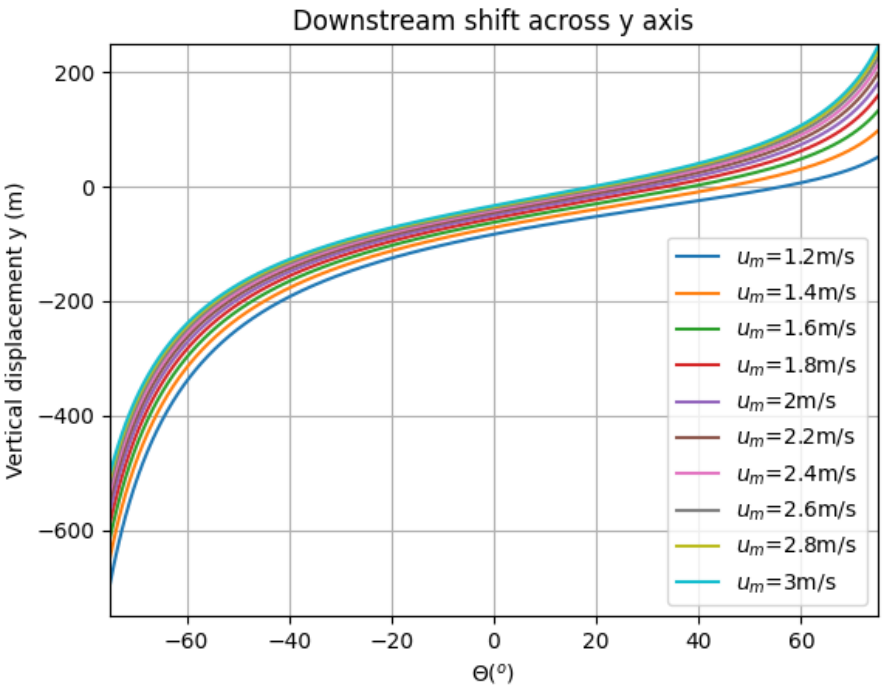


Figure 8: Y-axis deviation with each corresponding θ for 10 different values of the swimmer's constant relative speed u_m. The parameters are shown in Table 1, example 2.

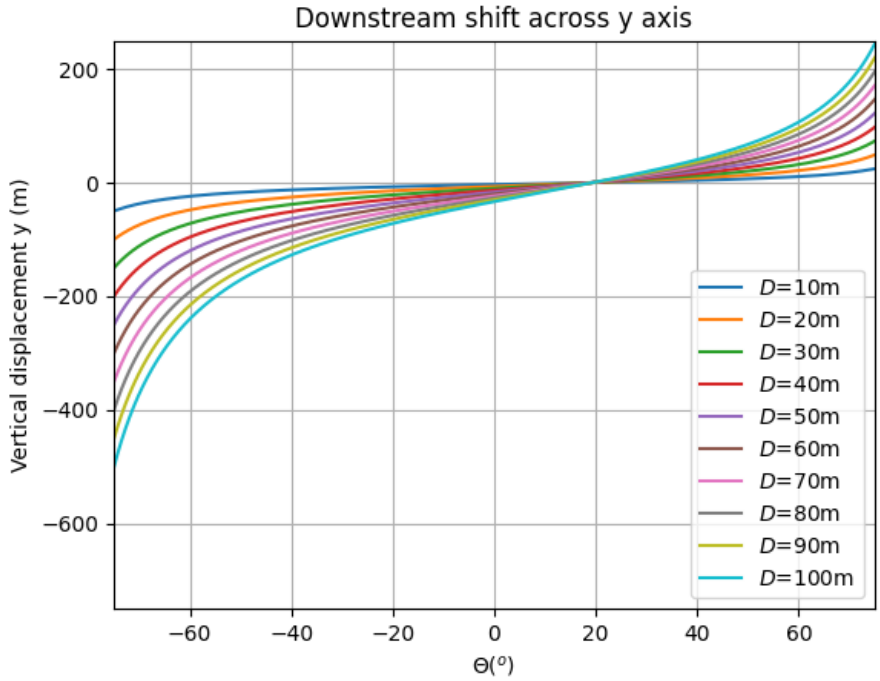


Figure 9: Y-axis deviation with each corresponding θ for 10 different values of the river's width D. The parameters are shown in Table 1, example 3.

3.2 Case of Constant River Flow and Decreasing Swimmer Speed

Several parametric simulations were conducted to validate the proposed methodology for the case of exponentially decreasing swimmer speed. The examined testcases, along with the selected

values of the exponential decay factor $\alpha>0$, the swimmer's initial relative speed u_{m0} and the river width D, are summarized in Table 2.

In the first test case, the influence of the exponential decay parameter α was investigated. The resulting terminal position coordinate y_B as a function of the swimming angle θ is presented in Fig. 10. For $\alpha=0$, corresponding to constant swimmer speed, the problem reduces to the previously studied constant-velocity case. As α increases, the swimmer's speed decreases more rapidly due to fatigue effects, requiring progressively larger swimming angles to compensate for the downstream drift and achieve the target condition $y_t=0$m. The results further indicate that for $\alpha \geqslant 0.05$, with $u_s=1$m/s, $u_{m0}=1.7$m/s, and $D \geqslant 10$m, the swimmer is unable to reach the desired target position, regardless of the selected trajectory relative angle θ.

In the second test case, the effect of the swimmer's initial relative speed u_{m0} was examined for fixed values of $\alpha=0.03$ and $u_s=1$m/s. The variation of the terminal coordinate y_B with respect to the swimming angle θ is illustrated in Fig. 11. The results show that, for a river width of D=10m, the minimum initial swimmer speed required to successfully reach the target position $y_t=0$m is $u_{m0}=1.48$m/s, achieved at an optimal swimming angle of $\theta^*=60^o$.

In the third test case, the influence of the river width D was analyzed for $u_s=1$m/s, $u_{m0}=1.7$m/s, and $\alpha=0.03$. The corresponding variation of y_B as a function of θ is depicted in Fig. 12. The results demonstrate that the maximum river width for which the swimmer can still reach the desired target position is D=16.5m, obtained for an optimal swimming angle of $\theta^*=55^o$. Beyond this threshold, the combined effects of downstream advection and swimmer fatigue prevent successful trajectory compensation.

Table 2: Parameters settled for uniform downstream and exponentially decreasing swimmer's speed case simulation examples.

Parameter	**Example 1**	**Example 2**	**Example 3**
River Width D in m	10	10	{5, 15, 15, 16.5}
River downstream u_s (m/s)	1	1	1
Swimmer's initial relative speed u_{m0} (m/s)	1.7	{1.3, 1.48, 1.5, 1.7, 2}	1.7
Exponential decrease α (eq1/s)	{0, 0.01, 0.03, 0.04, 0.05, 0.06}	0.03	0.03
Final Target Position y_t (m)	0	0	0

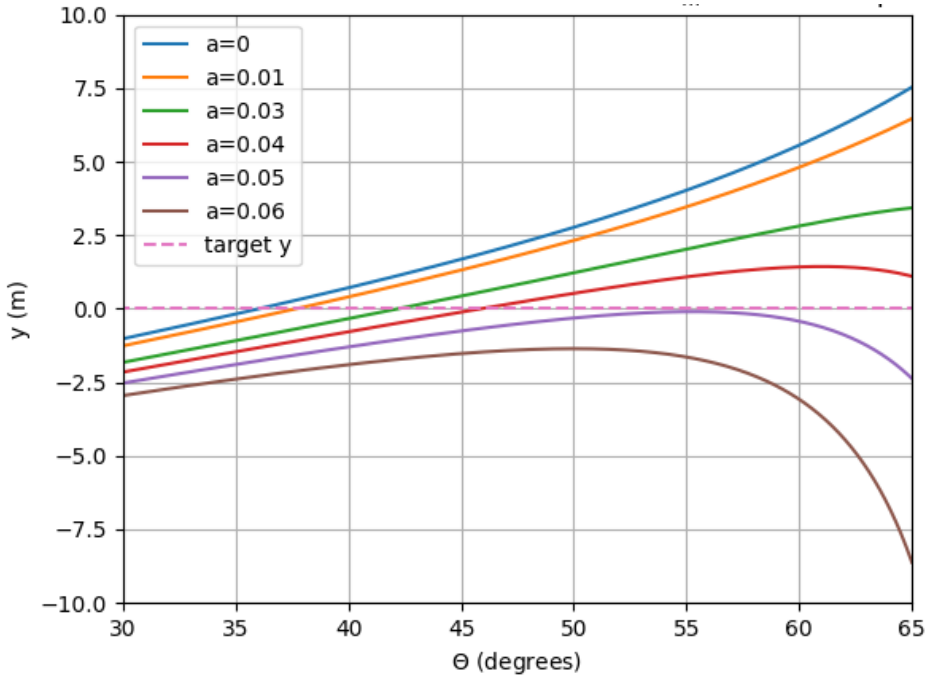


Figure 10: Y-axis deviation with each corresponding θ for 5 different values of α.

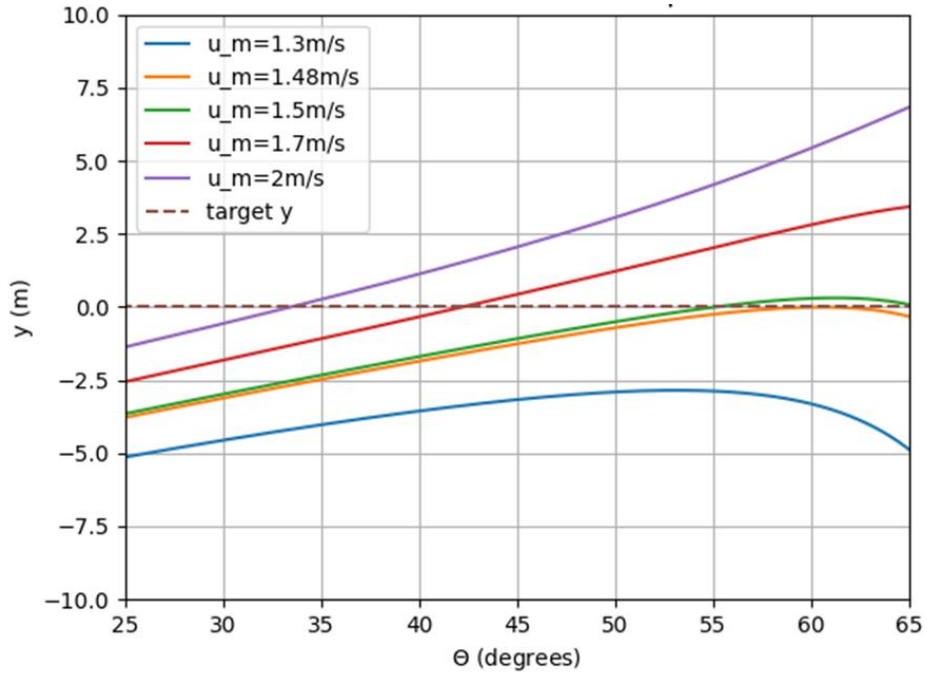


Figure 11: Y-axis deviation with each corresponding θ for 5 different values of the swimmer's initial relative speed u_{m0}.

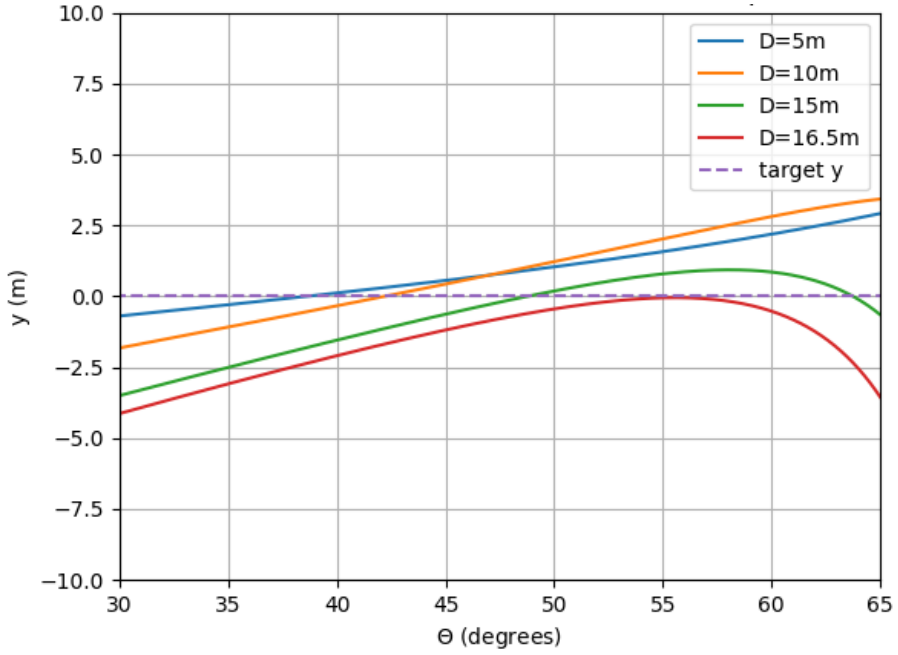


Figure 12: Y-axis deviation with each corresponding θ for 5 different values of the river's width D.

3. 3 Case of distributed river flow

Within this particular work, to validate the approach presented for the case of a spatially distributed river flow, a set of examples has been simulated, based on the parametric distribution described in (eq15). Therefore, the total velocity (eq14) in discretized form is written as:

$$\boldsymbol{u_t}(x_k, y_k, \theta_k) = \begin{bmatrix} u_m \cos\theta_k + \frac{Q}{2\pi}\left(\frac{x_k - x_{m1}}{R_{1,k}^2} - \frac{x_k - x_{m2}}{R_{2,k}^2}\right) \\ u_m \sin\theta_k + -V + \frac{Q}{2\pi}\left(\frac{y_k - y_{m1}}{R_{1,k}^2} - \frac{y_k - y_{m2}}{R_{2,k}^2}\right) \end{bmatrix} \text{(eq23)}$$

The parameters deployed for the simulation of this case, including the physical ones (D, u_m, u_s), the river stream ones (V, Q, flow centers), the target position and the GDA ones (initial guess of θ, number of timesteps N_p and the learning rate α) are included in Table 3. Three particular examples were studied. The first one assumes stronger offset downstream, but the whirl affects a smaller area. The swimmer's speed was heuristically set to a constant 1.45m/s while the total time to cross the river was t_d=10sec. In the other two examples the offset downstream V was halved; but the disturbance area coefficient Q was doubled. In both instances the total time and the swimmer's speed were heuristically set to 12.7 seconds and 1m/s respectively. However, the learning rate in the second example was double compared to the third, in order to observe its impact to the algorithms performance, under these defined physical parameters.

Table 3: Parameters for distributed river flow examples.

Parameter	Example 1	Example 2	Example 3
River Width D (m)	10	10	10
River downstream V (m/s)	1	0.5	0.5
Swimmer's relative speed u_m (m/s)	1.45	1	1
Disturbance Area coefficient Q (m^2/s)	1	2	2
Final Target Position y_t (m)	5.0	5.0	5.0
Learning Rate α	0.005	0.01	0.005
Total Time t_d (sec)	10	12.7	12.7
Time steps N_p	100	100	100
Initial θ (rad)	0.87	0.87	0.87
x_{m1} (m)	-1	-1	-1
y_{m1} (m)	1	1	1
x_{m2} (m)	1	1	1
y_{m2} (m)	-1	-1	-1

The corresponding results including the final paths and the GDA convergence rate curves are depicted in Figs 13- 19. As it is observed in Fig. 13, the swimmer's path angle θ is adapted to the river's stream in order to effectively cross it (zoomed-in Fig. 14). The cost per iteration graph in Fig. 15 shows that the deviation from the target is reduced to 10cm after a number of 200 iterations with a learning rate of $\alpha = 0.005$.

In addition, Figs. 16 and 18 depict a more significant deviation from the initial path the would be adopted with an absent flow disturbance since the disturbance coefficient is doubled while the swimmer's speed was reduced and the total time was increased.

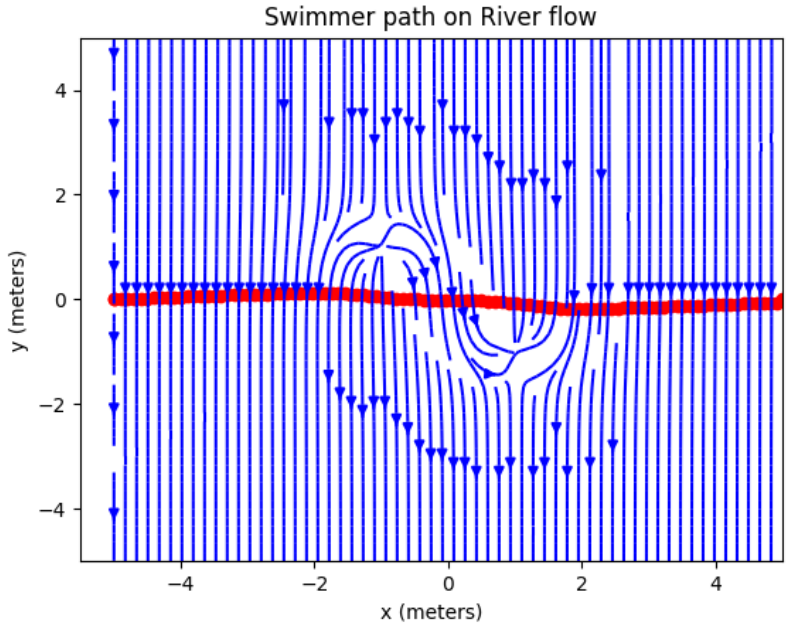


Figure 13: Optimal swimmer's path, after GDA execution for the parameters in example 1 (Table 3).

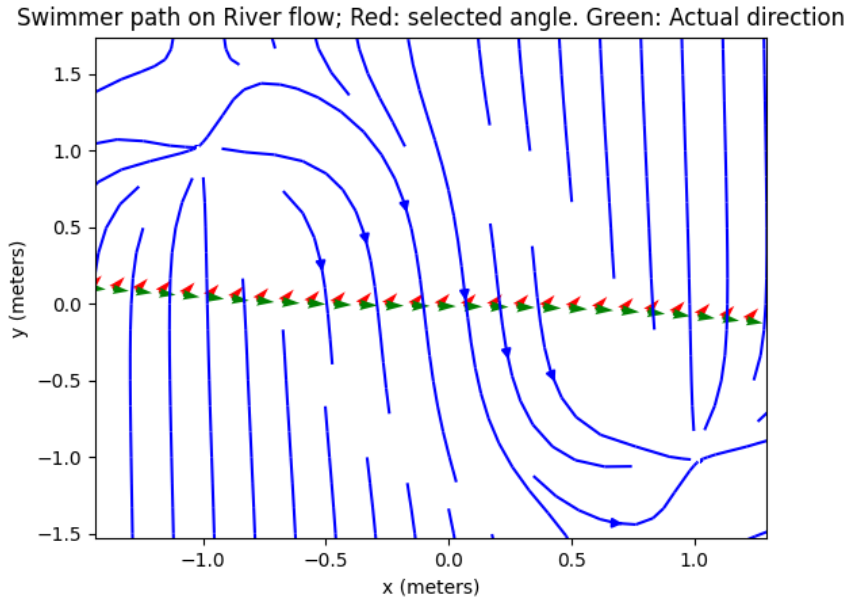


Figure 14: Zoomed detail of Fig. 13 that shows the swimmer's relative velocity angle (red arrows) and the total velocity's angle (green arrows).

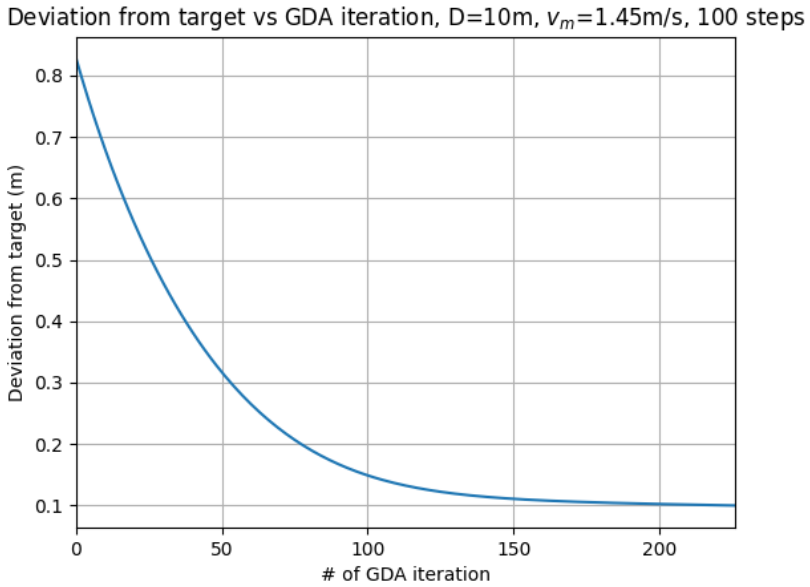


Figure 15: Convergence curve of the GDA execution for the parameters in example 1 (Table 3).

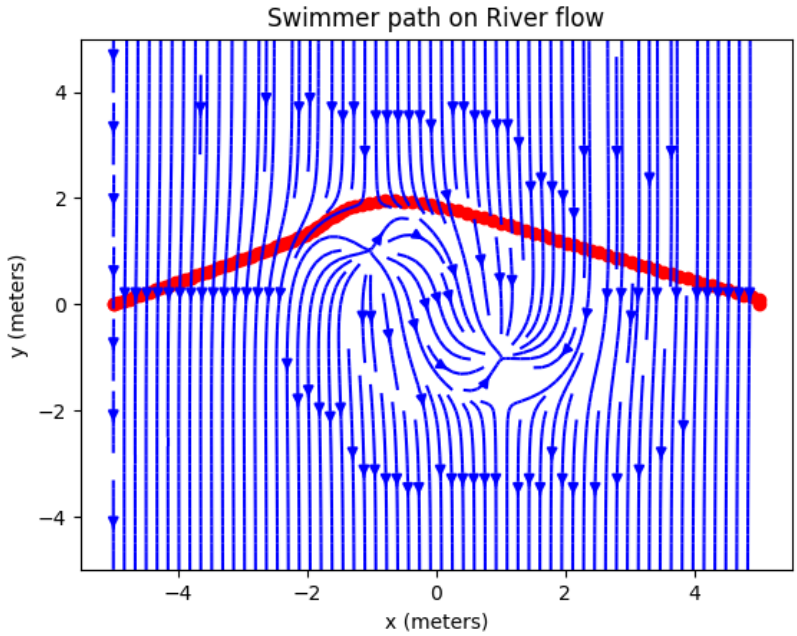


Figure 16: Optimal swimmer's path, after GDA execution for the parameters in example 2 (Table 3).

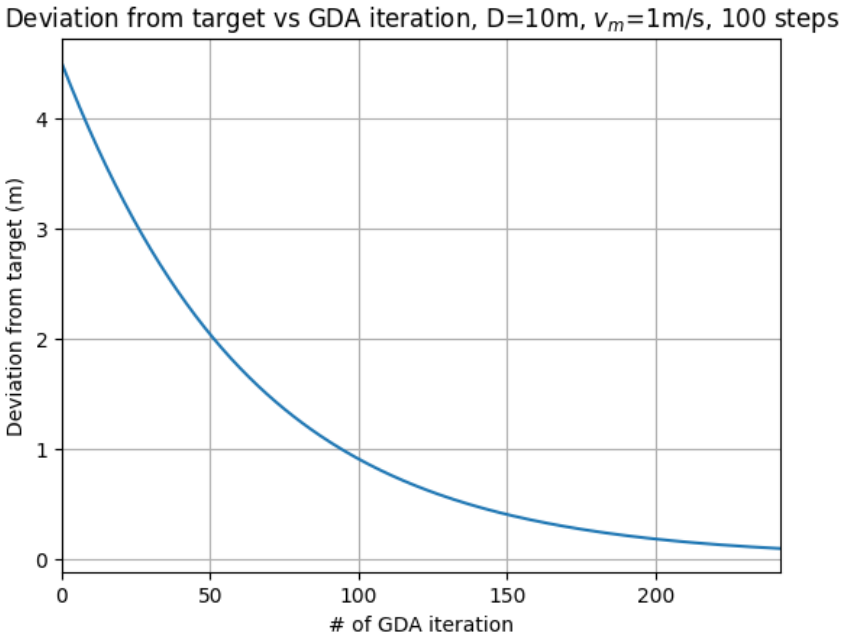


Figure 17: Convergence curve of the GDA execution for the parameters in example 2 (Table 3).

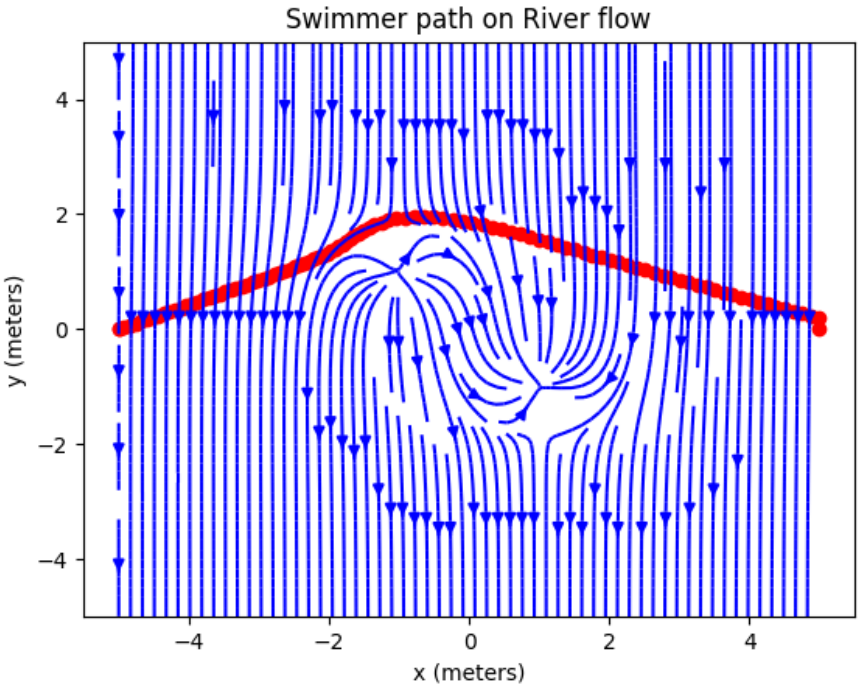


Figure 18: Optimal swimmer's path, after GDA execution for the parameters in example 3 (Table 3).

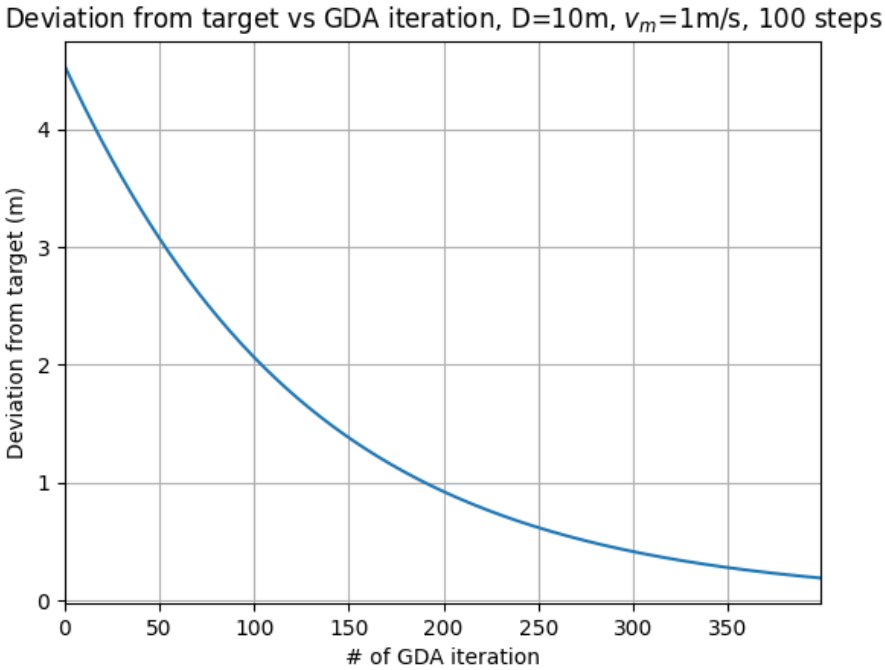


Figure 19: Convergence curve of the GDA execution for the parameters in example 3 (table 3).

3.4 Discussion

The results obtained across these case studies highlight the fundamental interplay between swimmer propulsion and flow-induced advection. For the uniform-flow scenarios, the optimal swimming direction is governed by the balance between the swimmer's transverse velocity

component and the downstream drift generated by the river current. The parametric analysis demonstrates that increases in river width or downstream velocity require larger upstream-oriented swimming angles to compensate for the accumulated lateral displacement. Conversely, increasing the swimmer's relative speed reduces the required relative swimming angle θ, since the crossing time t_d decreases and the swimmer is exposed to the downstream current flow for shorter time. These observations are consistent with the analytical expressions derived in Sections 2.1 and 2.2 and provide a physical interpretation of the obtained optimal trajectories.

The introduction of swimmer fatigue through an exponentially decreasing velocity, significantly alters the solution. As the decay factor α increases, the effective propulsion available during the latter stages of the crossing decreases, leading to longer crossing times t_d and larger vertical displacement deviations due to downstream. The existence of critical parameter thresholds, beyond which the target location becomes unreachable, illustrates the non-linear dependence of trajectory feasibility on the swimmer's endurance characteristics.

For the spatially distributed flow case, the trajectory optimization framework successfully adapts the swimmer's heading to local flow conditions. The adopted flow field, represented by a source–sink pair superimposed on a uniform downstream current, creates regions of the river that perturb the swimmer's nominal path. The GDA was able to identify angle sequences that exploit favorable flow regions while avoiding excessive drift, reducing the final position error. The convergence results further indicate that the optimization performance is sensitive to the learning rate selection, with larger values leading to faster convergence but potentially less stable optimization behavior. Although the finite difference and GDA framework proved effective for the scenarios considered, more complex flow environments may require advanced or learning-based optimization techniques to improve computational efficiency and robustness.

4. Conclusion

In this work, the problem of optimal trajectory planning for a swimmer crossing a river was investigated under several scenarios and parameter configurations. In particular, three cases were examined: a uniform river current with constant swimmer speed, a uniform current with exponentially decreasing swimmer speed and a non-uniform spatially distributed flow field. Key system parameters, including swimmer and river speed, river width, the exponential decay factor of the swimmer's speed and the spatial characteristics of the non-uniform flow, were systematically varied to analyze their influence on the resulting trajectory. To address these scenarios, physics, calculus, numerical and optimization approaches were combined, including, the finite difference method and a gradient-descent optimization algorithm. The results demonstrate that, under the adopted assumptions, the swimmer's optimal path can be effectively determined using numerical and optimization techniques. These findings highlight that relatively simple mathematical models can capture essential aspects of trajectory planning in flowing environments.

As future work, the framework could be extended to more complex and realistic scenarios, such as time-dependent and spatially varying flow fields, variable swimmer speed, non-constant river widths and the presence of obstacles. Addressing such cases requires more advanced optimization

and control approaches, such as learning-based algorithms or optimal control formulations, to ensure robust and stable trajectory planning across a wider range of environmental conditions.

Conflict of Interest

The author declares no conflicts of interest.

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5. References

1. Wang, Y., Hernandez, A., Shen, L., & Zhang, H. (2024). Path planning of water surface garbage cleaning robot based on improved immune particle swarm algorithm. *AIP Advances*, *14*(2). https://doi.org/10.1063/5.0181605

2. Dy, C. K., Lee, L. M., Lim, J. K., Lu Chiu, D. E., Santos, J., Chua, J. A. et al. (2025, May). Development of a path planning algorithm for a water unmanned surface vehicle using particle swarm optimization method for water quality monitoring. In IOP Conference Series: Earth and Environmental Science (Vol. 1500, No. 1, p. 012078). IOP Publishing. DOI: 10.1088/1755-1315/1500/1/012078

3. Wu, G., Wang, M., & Guo, L. (2024). Complete coverage path planning based on improved genetic algorithm for unmanned surface vehicle. *Journal of Marine Science and Engineering*, *12*(6), 1025. https://doi.org/10.3390/jmse12061025

4. Alageshan, J. K., Verma, A. K., Bec, J., & Pandit, R. (2020). Machine learning strategies for path-planning microswimmers in turbulent flows. Physical Review E, 101(4), 043110. https://doi.org/10.1103/PhysRevE.101.043110

5. Xu, Z., Liu, Y., Yao, H. D., Wang, S., & He, G. (2026). Onset of vortex shedding in flow past Rankine ovals. Ocean Engineering, 359, 125779. 10.1016/j.oceaneng.2026.125779

6. Kedam, N., Tiwari, D. K., Kumar, V., Khedher, K. M., & Salem, M. A. (2024). River stream flow prediction through advanced machine learning models for enhanced accuracy. *Results in Engineering*, *22*, 102215. https://doi.org/10.1016/j.rineng.2024.102215

7. Ziadi, S., Chokmani, K., Chaabani, C., & El Alem, A. (2024). Deep learning-based automatic river flow estimation using RADARSAT imagery. *Remote Sensing*, *16*(eq10), 1808. https://doi.org/10.3390/rs16101808

8. Nawaz, Y., Arif, M. S., Bibi, M., Abbasi, J. N., Javed, U., & Nazeer, A. (2020). A finite difference method and effective modification of gradient descent optimization algorithm for MHD fluid flow over a linearly stretching surface. *Computers, Materials, & Continua*, *62*(eq2), 657. DOI:10.32604/cmc.2020.08584

9. Suner, M., Salci, S. A., Yigit, K. S., & Kandemir, I. (2015). Analytical analysis of hydrodynamics of the perforated Rankine oval. Ocean Engineering, 108, 227-240. 10.1016/j.oceaneng.2015.07.018

10. Kaushik, M. (2018). Potential flow theory. In *Theoretical and Experimental Aerodynamics* (pp. 107-126). Singapore: Springer Singapore. https://doi.org/10.1007/978-981-13-1678-4_4

11. LeVeque, R. J. (1998). Finite difference methods for differential equations. *Draft version for use in AMath*, *585*(6), 112.

12. Shyy, W. (1985). A study of finite difference approximations to steady-state, convection-dominated flow problems. *Journal of Computational Physics*, *57*(3), 415-438. DOI: 10.1016/0021-9991(85)90188-3

13. Ruder, S. (2016). An overview of gradient descent optimization algorithms. *arXiv preprint arXiv:1609.04747*. https://doi.org/10.48550/arXiv.1609.04747

14. Haji, S. H., & Abdulazeez, A. M. (2021). Comparison of optimization techniques based on gradient descent algorithm: A review. *PalArch's Journal of Archaeology of Egypt/Egyptology*, *18*(4), 2715-2743.

15. Campana, M., Lamiraux, F., & Laumond, J. P. (2016). A gradient-based path optimization method for motion planning. Advanced Robotics, 30(17-18), 1126-1144. https://doi.org/10.1080/01691864.2016.1168317